Purely Analytic Ledger Replacement, Packet D: Seam Arithmetic and the Two-Sided Payments

Exact hand-checkable exhibit

Abstract

This packet replaces the twenty-two executable seam predicates and the eleven moving/fixed payment predicates used in the high side through k_6. Every premise is an analytic inequality already derived in the seam or staircase argument, and every remaining decision is displayed as an exact rational identity or strict rational comparison. In particular, the derivative of the seam comparison function is derived explicitly and is bounded by an exact proxy with a positive margin to 16/5. The six moving k_6 payments are printed with their full rational differences, not merely their numerators. No program execution, floating-point value, or interval calculation is a premise.

1 Rational constants used by both tables

For completeness, the needed bounds for π can be checked from the alternating arctangent series, rather than imported from an executable Machin enclosure. Machin's identity and the alternating remainder rule give

$$\pi > 16\left(\frac{1}{5} - \frac{1}{3\,5^3} + \frac{1}{5\,5^5} - \frac{1}{7\,5^7}\right) - \frac{4}{239} =: L_\pi,$$
$$\pi < 16\left(\frac{1}{5} - \frac{1}{3\,5^3} + \frac{1}{5\,5^5}\right) - 4\left(\frac{1}{239} - \frac{1}{3\,239^3}\right) =: U_\pi.$$

Indeed, the tangent-addition formula proves $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, and the displayed truncations have the indicated directions. Direct rational subtraction gives

$$L_\pi - \frac{333}{106} = \frac{3418213}{41563593750} > 0, \qquad \frac{1571}{500} - U_\pi = \frac{323354809}{853244937500} > 0. \tag{1}$$

Since

$$\frac{22}{7} - \frac{1571}{500} = \frac{3}{3500} > 0, \tag{2}$$

we may use throughout

$$\underline{\pi} := \frac{333}{106} < \pi < \overline{\pi} := \frac{1571}{500} < \frac{22}{7}. \tag{3}$$

All later occurrences of a rational π bound refer to (3).

2 The twenty-two seam rows

2.1 Analytic input and comparison function

We recall precisely the analytic data to which the finite arithmetic is applied. Put

$$\epsilon = 1 - \rho = y^2, \qquad \kappa = K\epsilon^2, \qquad 0 < y \le \frac{1}{\sqrt{6}}, \quad \rho \ge \frac{5}{6}, \quad \kappa \ge 54.$$

For a first plateau of length p, and the quantities m, R in the seam decomposition, set

$$P = py, \qquad r = Ry, \qquad S = (p+m)y.$$

The angular estimate $d > 5/8$ and $R = p - dm$ imply, whenever $r > 0$,

$$0 < r \le P, \quad S \le S_*(P,r) := \frac{13P - 8r}{5}. \tag{4}$$

Equality in the second relation can occur when $m = 0$; no later step requires strictness at this comparison. The curvature-shelf calculation already proved in the seam argument gives the self-consistent inequality

$$P^2 < \frac{2\pi^2 Q}{(1-\widehat{q})^2}, \quad Q = 1 + \frac{y^2}{\kappa} + \frac{yS}{\kappa}, \quad \widehat{q} = \frac{y^2}{\rho}(Q-1). \tag{5}$$

Since the right-hand side increases with Q, define

$$Q_* = 1 + \frac{y^2}{\kappa} + \frac{yS_*}{\kappa}, \quad q_* = a(Q_* - 1), \qquad a := \frac{y^2}{\rho} \le \frac{1}{5}, \tag{6}$$

and

$$H(P,r) := \frac{2\pi^2 Q_*}{(1-q_*)^2}. \tag{7}$$

Then (4) gives $Q \le Q_*$ and $\widehat{q} \le q_*$. Since (5) is itself strict and its right-hand side increases with Q, it follows that $P^2 < H(P,r)$.

The remaining arithmetic is exactly the following table. Rows 1–5 are the five constant/shelf rows, rows 6–10 are the five displacement rows, rows 11–14 are the four derivative rows, rows 15–17 are the three endpoint rows, and rows 18–22 are the five terminal-payment rows. Thus the table reconciles one-for-one with all twenty-two seam predicates.

For clarity, the radical used in rows 7–9 follows directly from the analytic shelf bound. Namely,

$$p^2 < \frac{2\pi\rho K}{\epsilon}, \quad P = py, \quad K = \frac{\kappa}{y^4}$$

imply

$$\left(\frac{y^3 P}{\kappa\rho}\right)^2 < \frac{2\pi y^2}{\kappa\rho} \le \frac{\pi}{135} < \frac{\overline{\pi}}{135}.$$

Together with $S_* \le 13P/5$, this gives

$$q_* < \frac{1}{1620} + \frac{13}{5}\frac{763}{5000},$$

which is precisely the dependency recorded by rows 6–9.

#	quantity	exact rational check	role
1	seam π bound	$\overline{\pi} - U_\pi = 323354809/853244937500 > 0$	$\pi < \overline{\pi}$; (1)
2	lower $\sqrt{2}$ bound	$2 - (140/99)^2 = 2/9801 > 0$	$140/99 < \sqrt{2}$
3	upper $\sqrt{2}$ bound	$(99/70)^2 - 2 = 1/4900 > 0$	$\sqrt{2} < 99/70$
4	y ceiling	$(49/120)^2 - 1/6 = 1/14400 > 0$	$y < 49/120$
5	shelf ceiling	$(13/5)^2 - 2\overline{\pi} = 119/250 > 0$	$\sqrt{2\pi\rho} < 13/5$; rows 1

#	quantity	exact rational check	role
6	first displacement term	$\frac{(1/6)^2}{54(5/6)} = \frac{1}{1620}$	$y^4/(\kappa\rho) \le 1/1620$
7	radical ceiling	$(763/5000)^2 - \frac{\overline{\pi}}{135} = \frac{8563}{675000000} > 0$	shelf radical; row 1
8	displacement proxy	$\frac{1}{1620} + \frac{13}{5}\frac{763}{5000} = \frac{804689}{2025000}$	rows 6–7
9	displacement margin	$\frac{159}{400} - \frac{804689}{2025000} = \frac{497}{4050000} > 0$	$q_*, \widehat{q} < \overline{q} := 159/400$
10	outer-half localization	$\frac{5}{6}(1 - 159/400) = \frac{241}{480} = \frac{1}{2} + \frac{1}{480}$	$x_0/K > 1/2$; row 9
11	Q_*-derivative proxy	$D_Q := \frac{13}{5}\frac{49/120}{54} = \frac{637}{32400}$	$\partial_P Q_* < D_Q$; row 4
12	exact H-derivative proxy	$\frac{2\overline{\pi}^2 D_Q(1 + 159/400 + 2/5)}{(1 - 159/400)^3} = \frac{2260740364246}{708624500625}$	rows 1, 9, 11 and $a \le 1/5$
13	margin to 16/5	$\frac{16}{5} - \frac{2260740364246}{708624500625} = \frac{6858037754}{708624500625} > 0$	$\partial_P H(P, B) < 16/5$
14	synthetic-slope margin	$2\frac{14}{3} - \frac{16}{5} = \frac{92}{15} > 0$	$\partial_P(P^2 - H) > 92/15$
15	endpoint Q_* proxy	$\overline{Q} := 1 + \frac{1/6}{54} + \frac{(49/120)(14/3)}{54} = \frac{10093}{9720}, \quad \overline{Q} - 1 = \frac{373}{9720}$	$Q_*(B, B) < \overline{Q}$
16	endpoint q_* proxy	$\frac{1}{5}(\overline{Q} - 1) = \frac{373}{48600}$	$q_*(B, B) < 373/48600$
17	initial reserve	$\left(\frac{14}{3}\right)^2 - \frac{2\overline{\pi}^2(10093/9720)}{(1 - 373/48600)^2} = \frac{2505132463469}{2616573970125} > 0$	$B^2 - H(B, B) > 0$; rows 1, 15–16
18	action-gain coefficient	$g := \frac{2(140/99)54}{3(1571/500)} = \frac{280000}{17281}$	$K\eta > g/y$; rows 1–2
19	gain after R loss	$g - \frac{14}{3} = \frac{598066}{51843}$	use $R < 14/(3y)$
20	interface loss proxy	$\frac{8}{15}\frac{99}{70} = \frac{132}{175}$	$4\delta < 132/175$; row 3
21	dangerous-branch reserve	$\frac{598066}{51843}\frac{120}{49} - 1 - \frac{132}{175} = \frac{80132733}{3024175} > 0$	$R > 0$; rows 4, 19–20
22	safe-branch reserve	$\frac{280000}{17281}\frac{120}{49} - 1 - \frac{132}{175} = \frac{114694733}{3024175} > 0$	$R \le 0$; rows 4, 18, 20

2.2 Why the table proves the seam implications

We spell out the two implications for which the table is more than a collection of arithmetic facts.

First, on the segment $r = B := 14/3$ and $P \ge B$,

$$\partial_P Q_* = \frac{13y}{5\kappa}, \quad \partial_P q_* = a\,\partial_P Q_*.$$

Differentiation of (7), followed only by the identity $aQ_* = q_* + a$, gives

$$\partial_P H = 2\pi^2(\partial_P Q_*)\frac{1 - q_* + 2aQ_*}{(1-q_*)^3} = 2\pi^2(\partial_P Q_*)\frac{1 + q_* + 2a}{(1-q_*)^3}. \tag{8}$$

Rows 1, 9, and 11 and $a \le 1/5$ substitute monotonically into (8); rows 12–13 then prove $\partial_P H < 16/5$. Because $P \ge B$, row 14 proves

$$\frac{d}{dP}(P^2 - H(P,B)) = 2P - \partial_P H(P,B) > \frac{92}{15} > 0. \tag{9}$$

Rows 15–17 prove $B^2 - H(B,B) > 0$. Integration of (9) therefore yields $P^2 > H(P,B)$ for every $P \ge B$. If $r \ge B$, then $H(P,r) \le H(P,B)$ because $S_*(P,r)$ decreases with r and H increases with Q_*. This contradicts $P^2 < H(P,r)$. Hence

$$r < B, \quad R < \frac{14}{3y}. \tag{10}$$

Second, the turning-point action bound gives

$$K\eta \ge \frac{2\sqrt{2}\,\kappa}{3\pi}\frac{1}{y} > \frac{g}{y},$$

where row 18 is the exact rational lower proxy. Also row 4 gives $1/y > 120/49$, and row 20 gives $4\delta < 8\sqrt{2}/15 < 132/175$. With $M = \lfloor K\eta \rfloor - R$ and the strict inequality $\lfloor x \rfloor > x - 1$, (10) and row 21 give, when $R > 0$,

$$M - 4\delta > \left(g - \frac{14}{3}\right)\frac{120}{49} - 1 - \frac{132}{175} > 0.$$

When $R \le 0$, row 22 gives the still stronger estimate

$$M - 4\delta > g\frac{120}{49} - 1 - \frac{132}{175} > 0.$$

Thus the seam tail has the strict action payment claimed in the main argument, including the $R = 0$ and integer-$K\eta$ faces.

Remark 1 (Notation correction in the executable label). The supplemental verifier labels row 11 as an "exact dq/dP cap." In the notation of the proof, the number 637/32400 is a proxy for $\partial_P Q_*$, not for $\partial_P q_*$. In fact $\partial_P q_* = a\partial_P Q_*$ and hence $\partial_P q_* < 637/162000$. Formula (8) and the verifier both use 637/32400 in its correct role as the Q_*-derivative proxy. The proof is therefore sound, but the old ledger label is inaccurate.

Remark 2 (Source wording for the π bound). One sentence in the seam passage groups $\pi < 1571/500$ with bounds said to be proved by positive rational squaring. Squaring proves rows 2–5, but not a bound for π. Equations (1)– (3) supply the required analytic alternating-series proof.

3 The eleven k_6 payment rows

Put

$$z_\rho = \frac{\pi}{1-\rho}, \quad k_m(\rho)^2 = z_\rho^2 + m(m+1), \quad p_1(\rho)^2 = 4z_\rho^2 + 2, \tag{11}$$

and

$$\mathcal{W}(\rho, K) = \frac{2}{9\pi}(1-\rho^3)K^3. \tag{12}$$

For a moving wall $T^2 = az_\rho^2 + b$, direct differentiation shows that $\mathcal{W}(\rho, T)$ increases whenever

$$a\pi^2(1+\rho) > b\rho^2(1-\rho)^2. \tag{13}$$

Here $b/a \le 42$. Since $\rho^2(1-\rho)^2 \le 1/16$ and $\pi > 3$, the left side of (13), after division by a, is greater than 9, whereas the right side is at most 42/16; hence every moving payment in the table is increasing. At a fixed wall L, the function $\mathcal{W}(\rho, L)$ is decreasing in ρ.

For a rational split r, define

$$A(a,b;r) := a\frac{(333/106)^2}{(1-r)^2} + b, \quad B(r) := \frac{7}{99}(1-r^3). \tag{14}$$

The bounds (3) give, on a moving wall at $\rho = r$,

$$\mathcal{W}(r,T) > B(r)A(a,b;r)^{3/2}. \tag{15}$$

Thus, for a positive cap C, the rational inequality

$$\Delta_{\text{mov}} := B(r)^2 A(a,b;r)^3 - C^2 > 0 \tag{16}$$

implies the strict payment, because

$$\Delta_{\text{mov}} = \big(B(r)A^{3/2} - C\big)\big(B(r)A^{3/2} + C\big)$$

and the second factor is positive. At a fixed wall L, it is enough to check the ordinary rational reserve

$$\Delta_{\text{fix}} := B(r)L^3 - C > 0. \tag{17}$$

The following is the complete eleven-row registry: six moving rows and five fixed rows. The fractions in the last column are the full reduced differences in (16) or (17).

wall/branch	split r	wall data	cap C	full exact rational difference
k_1 moving	1/8	$(a,b) = (1,2)$	4	$\Delta_{\text{mov}} = \frac{153864824754340538785}{34879610119261785292 8} > 0$
k_2 moving	3/10	$(a,b) = (1,6)$	9	$\Delta_{\text{mov}} = \frac{1399810848073457853053}{394237491401523000000} > 0$
k_3 moving	1/2	$(a,b) = (1,12)$	16	$\Delta_{\text{mov}} = \frac{1370275053537366631}{514922437748928} > 0$
k_4 moving	1/2	$(a,b) = (1,20)$	25	$\Delta_{\text{mov}} = \frac{2507119282010251109}{13902905819221056} > 0$
k_5 moving	13/25	$(a,b) = (1,30)$	36	$\Delta_{\text{mov}} = \frac{92672404686738742434741}{709259556128000000000} > 0$
p_1 moving	1/5	$(a,b) = (4,2)$	29	$\Delta_{\text{mov}} = \frac{3732557720374789930028 33}{868931613701316000000} > 0$
k_2 fixed	3/10	$L = 51/10$	9	$\Delta_{\text{fix}} = \frac{1387329}{11000000} > 0$
k_3 fixed	1/2	$L = 13/2$	16	$\Delta_{\text{fix}} = \frac{6277}{6336} > 0$
k_4 fixed	1/2	$L = 15/2$	25	$\Delta_{\text{fix}} = \frac{775}{704} > 0$
k_5 fixed	13/25	$L = 17/2$	36	$\Delta_{\text{fix}} = \frac{452843}{343750} > 0$
p_1 fixed	1/5	$L = 77/10$	29	$\Delta_{\text{fix}} = \frac{849901}{281250} > 0$

We now verify that the table pays every branch rather than merely listing positive numbers. For each maximum entry wall

$$\max\{k_2, 51/10\}, \quad \max\{k_3, 13/2\}, \quad \max\{k_4, 15/2\}, \quad \max\{k_5, 17/2\}, \quad \max\{p_1, 77/10\},$$

partition the ratio interval at the displayed r. On $\rho \ge r$, the maximum is at least the moving component and monotonicity together with the moving row pays C. On $\rho \le r$, the maximum is at least the fixed component and the fixed row pays C. This argument does not assume that the two components are equal at the rational split. For k_1, the applicable ratio interval starts at $\rho_c > 1/8$ (because $\pi < 22/7 < 7/2$), so the first moving row applies directly. Strict payments also cover coincident entry walls, while the wall itself remains assigned to the lower cap by the strict spectral convention.

4 Reconciliation with the lower-d rows

The two rational π predicates used by the two-sided ledger are already proved in (3). The eleven k_6 predicates are exactly the eleven rows in the preceding table. The remaining thirteen wall/cap predicates in that ledger are the ten transparent multiplicity sums

$$\begin{aligned}&(1,\ 1+3,\ 1+3+5,\ 1+3+5+1,\ 1+3+5+7,\\ &\quad 1+3+5+7+1,\ 1+3+5+7+9+1,\\ &\quad 1+3+5+7+9+1+3,\\ &\quad 1+3+5+7+9+11+1+3,\\ &\quad 1+3+5+7+9+11+1+3+5)\\ &\qquad\qquad\qquad\qquad = (1,4,9,10,16,17,26,29,40,45),\end{aligned} \tag{18}$$

and the endpoint ordering $13/2 < 2z_\rho < 15/2$. The latter follows without an executable check: using $\sqrt{337} < 19$ on the lower endpoint and (3),

$$2\frac{333}{106}\frac{19}{18} - \frac{13}{2} = \frac{7}{53} > 0, \qquad \frac{15}{2} - \left(2\frac{22}{7} + 1\right) = \frac{3}{14} > 0. \tag{19}$$

Thus the cap–10 row is empty, and the three executable ordering rows collapse to (19) and its immediate empty-cell consequence.

The nine lower-d payment rows need no new table here: the main paper's lemma "Lower side through d" (label `cm:lem-lower-d`) already prints the seven fixed-wall differences

$$\frac{793292}{1546875}, \frac{486236661}{1375000000}, \frac{333100003}{99000000}, \frac{329797}{88000}, \frac{3590476297}{1125000000}, \frac{327078887}{99000000}, \frac{8805519}{1375000},$$

and states the initial and moving payments $\mathcal{W}(\rho_c, L(\rho_c)) > 19/6$, where $L(\rho) = (2\rho)^{-1}$ on this branch, and $\mathcal{W}(1/19, 2z) > 508/27$. If desired, their exact printed lower proxies close by the two one-line subtractions

$$\frac{57421}{16854} - \frac{19}{6} = \frac{675}{2809} > 0, \qquad \frac{173863}{8427} - \frac{508}{27} = \frac{137795}{75843} > 0. \tag{20}$$

Consequently every seam and two-sided payment predicate assigned to Packet D is now visible as analytic reasoning plus printed integer arithmetic.

5 Dependency count and conclusion

The replacement count is exact:

$$\underbrace{22}_{\text{seam table}} + \underbrace{(2+6+5+10+3)}_{k_6/\text{lower-}d\text{ wall–cap ledger}} + \underbrace{9}_{\text{already printed lower-}d\text{ payments}}.$$

The two constant rows are supplied by (3); the six plus five payment rows are the eleven-row table; the ten bookkeeping rows are (18); the three ordering rows are (19); and the nine already printed payments are explicitly cross-referenced above. This removes program execution from the logical dependency of precisely the Packet-D portion of the exact ledger.